

Single Source Justification (Simplified Acquisition)

Section A: General Contract Information

Is this a Bridge Action as defined in the [AF Bridge Action Reduction Plan](#)? ☐ Yes ☒ No

Contracting Activity: AFLCMC/PZIBB

Project/Program Name: Trace Gas Impurities Gas Chromatography System

Purchase Request / Local ID Number:

Estimated Contract Cost (including Options): \$127,222.00

Section B: Description of the Supplies/Services Required

AFPET/PTPLE has a requirement for one Trace Gas Impurities Gas Chromatography System to perform quantification of trace permanent gas impurities in Propellant-Grade Nitrogen, Helium, and Hydrogen. This GC system will be used to satisfy specification testing requirements for MIL-PRF-27401H (Nitrogen), MIL-PRF-27407E (Helium) and MIL-PRF-27201E (Hydrogen). This instrument will be used to perform specification testing on gases for quality assessment and monitoring of DoD/DLA product, as well as propellant pressurizing gases for use in space launch activity.

Section C: Justification for Soliciting from a Single Source or Brand Name

The United States Air Force has determined Custom Solutions Group LLC is the only known vendor that can build & provide an instrument that can detect and measure trace hydrogen, neon, argon, oxygen, helium and nitrogen that may be present in three separate gas products: high purity helium, high purity nitrogen and high purity hydrogen. The Custom Solutions Group system build accomplishes analysis in a single sample introduction & test event and meet the following salient characteristics: Instrument shall be fully compliant and provide test results in accordance with standard test method(s) described in MIL-PRF-27407E Propellant Pressurizing Agent Helium, MIL-PRF-27401H Propellant Pressurizing Agent Nitrogen and MIL-PRF-27201E Propellant Pressurizing Agent Hydrogen. Instrument shall be capable of quantifying trace Hydrogen, Argon, Oxygen, Neon and Nitrogen in bulk Helium gas IAW para 4.4.2 of MIL-PRF-27407E.

Section D: Efforts to Obtain Competition

Market research was conducted by interviews with government laboratories. Experiences at other laboratories indicate that units with single Pulsed Discharge Helium Ionization Detector (PDHID) cannot reliably detect and quantify traces of neon in bulk helium. **Custom Solutions Group LLC** was the only company that was aware of the unique problems associated with quantifying neon and had a technical solution for the reliable measurement of neon. Experiences at other laboratories indicates that instruments with thermal conductivity detectors that have an internal volume greater than 3.5 micro-liter cannot reliably detect and quantify helium in bulk hydrogen. All possible research has been exhausted and no other sources will suffice. **Custom Solutions Group LLC** is the only known source that has demonstrated the technical skill needed to incorporate all the capabilities described in the SOW into one instrument.

Section E: Steps to Preclude Future Single Source or Brand Name Awards

Future acquisitions of a suitable Trace Gas Impurities Gas Chromatography System will include a market survey to determine potential sources of suppliers of instruments that meet the requirement of trace gas analysis, specifically MIL-PRF-27407E, MIL-PRF-27201E, MIL-PRF-27401H. Currently, Custom Solutions Group, LLC is the only manufacturer and they do not have authorized distributor. Future acquisitions will also verify that instrument offered can test trace contaminants in helium, nitrogen, and hydrogen per the

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above mentioned military specifications.

Section F: Contracting Officer's Determination

I have determined, in accordance with [FAR 13.106-1\(b\)\(1\)](#), that the circumstances of this contract action deem only a single source is reasonably available.

Date	Typed Name and Rank/Grade of Contracting Officer	Signature of Contracting Officer
18-Mar-2026	Blackford, Jennifer	